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#include <iostream>
using namespace std;

char board[3][3] = {
    {'1', '2', '3'},
    {'4', '5', '6'},
    {'7', '8', '9'}
};

void displayBoard()
{
    cout << "\n";
    cout << "   |   |   \n";
    cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << "\n";
    cout << "_____|_____|_____\n";
    cout << "   |   |   \n";
    cout << " " << board[1][0] << " | " << board[1][1] << " | " << board[1][2] << "\n";
    cout << "_____|_____|_____\n";
    cout << "   |   |   \n";
    cout << " " << board[2][0] << " | " << board[2][1] << " | " << board[2][2] << "\n";
    cout << "   |   |   \n";
}

bool checkWin(char player)
{
    for (int i = 0; i < 3; i++)
    {
        if (board[i][0] == player &&
            board[i][1] == player &&
            board[i][2] == player)
            return true;

        if (board[0][i] == player &&
            board[1][i] == player &&
            board[2][i] == player)
            return true;
    }

    if (board[0][0] == player &&
        board[1][1] == player &&
        board[2][2] == player)
        return true;

    if (board[0][2] == player &&
        board[1][1] == player &&
        board[2][0] == player)
        return true;
}

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    return false;
}

bool checkDraw()
{
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            if (board[i][j] != 'X' && board[i][j] != 'O')
                return false;
        }
    }

    return true;
}

void resetBoard()
{
    char number = '1';

    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            board[i][j] = number++;
        }
    }
}

int main()
{
    char player = 'X';
    int choice;
    char again;

    cout << "=====\n";
    cout << "    TIC TAC TOE\n";
    cout << "=====\n";

    do
    {
        resetBoard();
        player = 'X';

        while (true)
        {
            displayBoard();

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cout << "\nPlayer " << player << ", enter position (1-9): ";
cin >> choice;

if (choice < 1 || choice > 9)
{
    cout << "Invalid position! Try again.\n";
    continue;
}

int row = (choice - 1) / 3;
int col = (choice - 1) % 3;

if (board[row][col] == 'X' || board[row][col] == 'O')
{
    cout << "Position already taken! Try again.\n";
    continue;
}

board[row][col] = player;

if (checkWin(player))
{
    displayBoard();
    cout << "\nPlayer " << player << " wins!\n";
    break;
}

if (checkDraw())
{
    displayBoard();
    cout << "\nGame Draw!\n";
    break;
}

if (player == 'X')
    player = 'O';
else
    player = 'X';
}

cout << "\nDo you want to play again? (y/n): ";
cin >> again;

} while (again == 'y' || again == 'Y');

cout << "\nThanks for playing!\n";

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    return 0;  
}
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